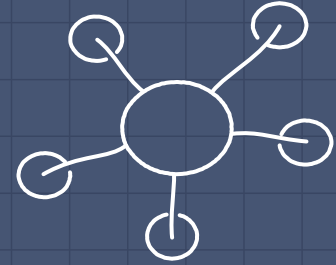


DATA SCIENCE PROJECT



Customer Churn Model

Delivered by Gerald Fraser for Pairview Ltd



AdventureWorks Hardware Group

Sales Performance
Dashboard

Recommendation
Engine

Churn Predictive
Model

Executive Summary



The 3 tools built in this project will empower the management of AHG to make smarter and more effective data driven decisions, and ultimately increase the commercial value of the business

The **Sales Performance Dashboard** will simplify and visualize KPIs and metrics in a summarized view which allows the business to make easier and quicker decisions

Through effective Product suggestions, the **Recommendation Engine** will incentivize customers to purchase more products and drive more revenue generated through online sales

The **Churn Predictive Model** is estimated to cut loss due to customer churn by up to **\$2m** every year



1. Sales Performance Dashboard

- Purpose of the dashboard
- Dashboard features – Filters & KPI's
- Dashboard screenshot



2. Recommendation Engine

- Need for a recommendation engine
- Data transformation, model training, evaluation
- Model Output example



3. Predictive Churn Model

- Purpose of a predictive model
- Data Exploration & RFM Customer Segmentation
- Feature Selection, Model Selection and Evaluation
- Commercial impact of churn / Recommendations to prevent churn

3. CHURN PREDICTIVE MODEL





3. CHURN PREDICTIVE MODEL

Purpose of Churn Model

- ❑ When a past customer has not made a purchase with the company for an extended period of time, that is classified as Customer Churn. The effect of customer churn can lead to reduced profitability of a business if not monitored effectively.
- ❑ AHG Management would require an efficient and quantifiable predictive model to conduct meaningful customer churn analytics to maintain customer retention
- ❑ Of AHG's 18,484 customer database - 5,511 customers have not made a purchase in the last 8 months. 8 months will be considered the time until customer churn for the business



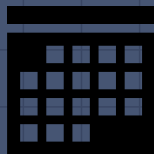


3. CHURN PREDICTIVE MODEL

Data Exploration and Transformation

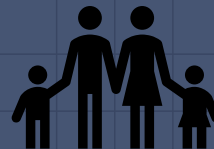
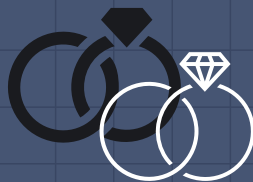
- ❑ Transactional, Demographic and Attitudinal data used
- ❑ Each data was aggregated converted/grouped by each unique customer and all the fields.
- ❑ The 3 datasets are merged to create 1 large dataframe showing single-view customer records

Transactional Data



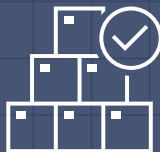
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Demographic Data



...

Attitudinal Data



...



3. CHURN PREDICTIVE MODEL

Data Exploration and Transformation

Further Info

- ❑ Revenue, Cost, Profit and Order Quantity were aggregated by min, max, mean and median for single customer view.
- ❑ Sales order reason and type were “hot-encoded” as customers would have more than one sales reason for buying items. Additionally, missing sales reason data was replaced with the mode.
- ❑ A new field **“Tenure_Months”**, which refers to the duration of customer’s first transaction date to their last transaction date, was also concatenated to the single-view dataframe. The **“Churn”** field relating to days since last transaction (or recency) was also added. [where churn=1 if recency>243 days]

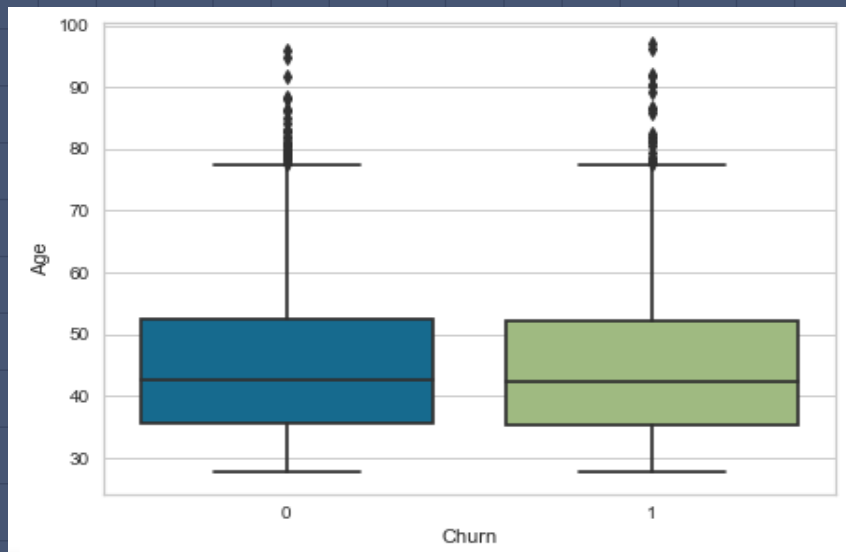




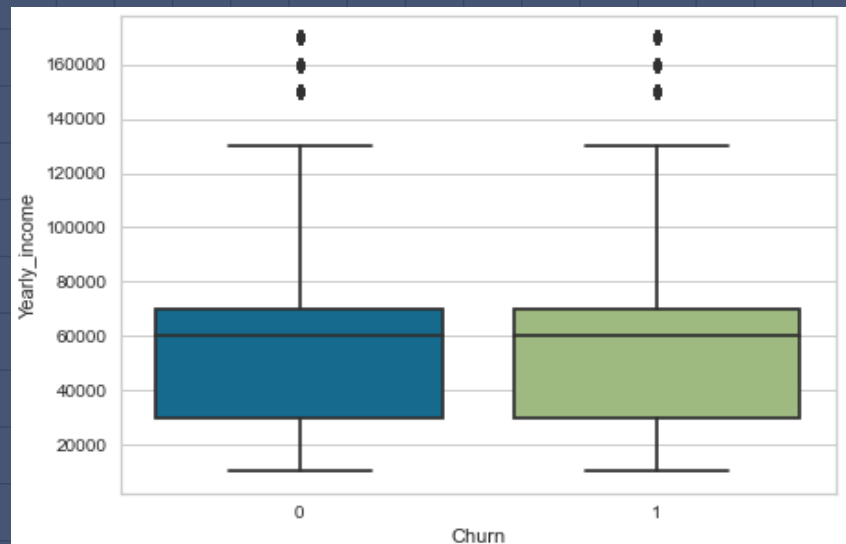
3. CHURN PREDICTIVE MODEL

Data Exploration Analysis

Churn by Age



Churn by Yearly Income



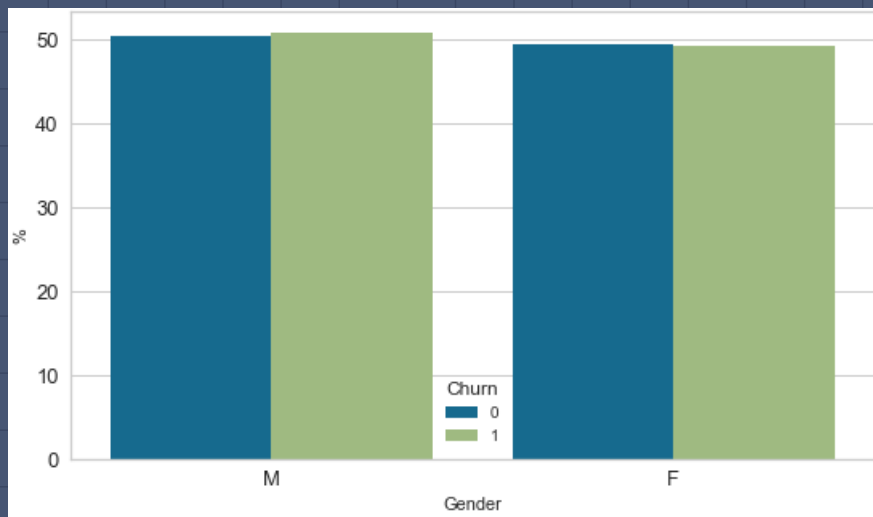
- ❑ There is virtually no variance between churners and non-churners in terms of Age and Yearly Income distributions among the customers

3. CHURN PREDICTIVE MODEL

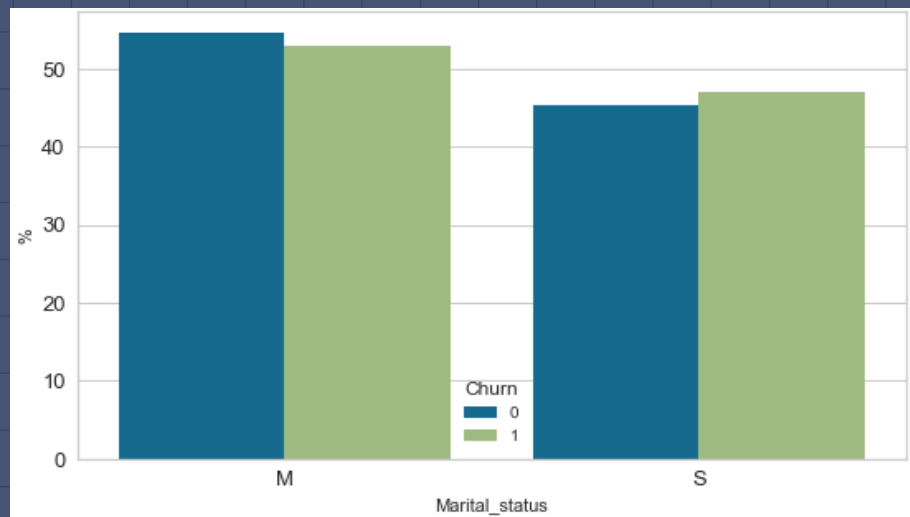


Data Exploration Analysis

Churn by Gender



Churn by Marital Status



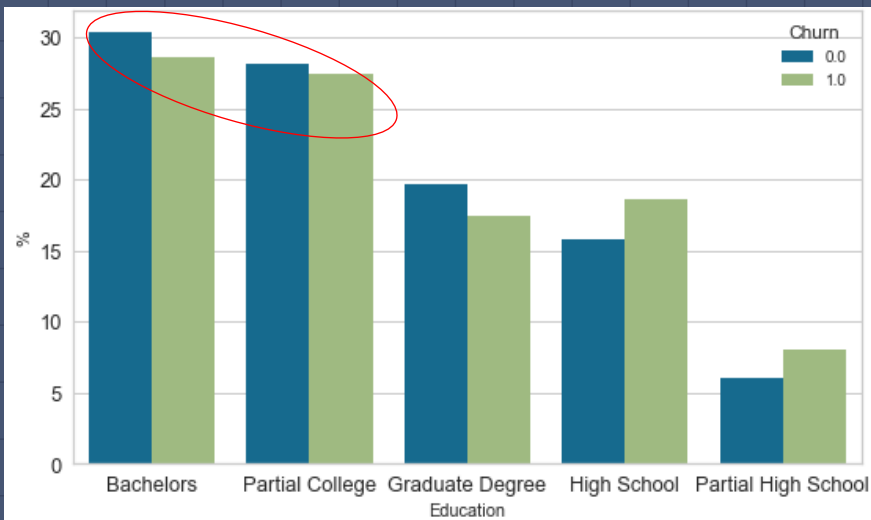
- ❑ There is virtually no variance between churners and non-churners in for Gender
- ❑ There is also virtually no variance for Marital Status among the Customers as well

3. CHURN PREDICTIVE MODEL



Data Exploration Analysis

Churn by Education



Churn by Occupation



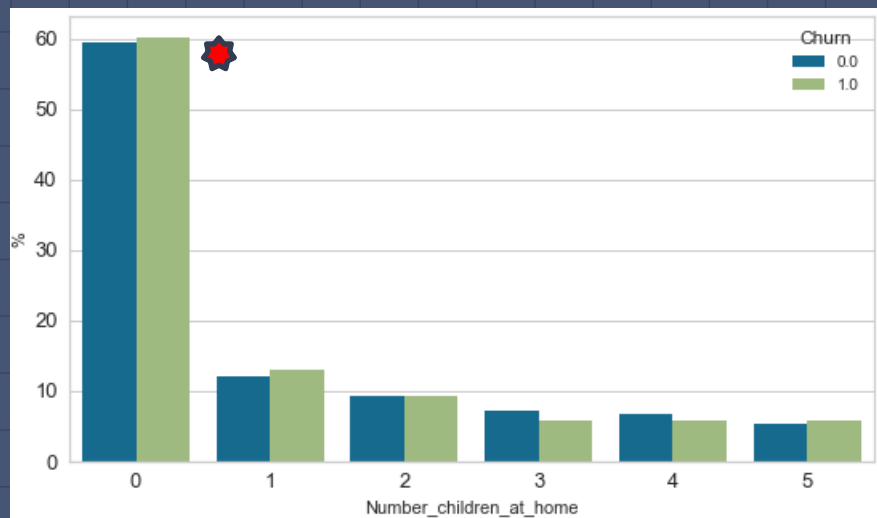
- ☐ **Bachelors** and **Partial College** customers are most likely to churn
- ☐ **Professional** and **Skilled Manual** customers are most likely to churn



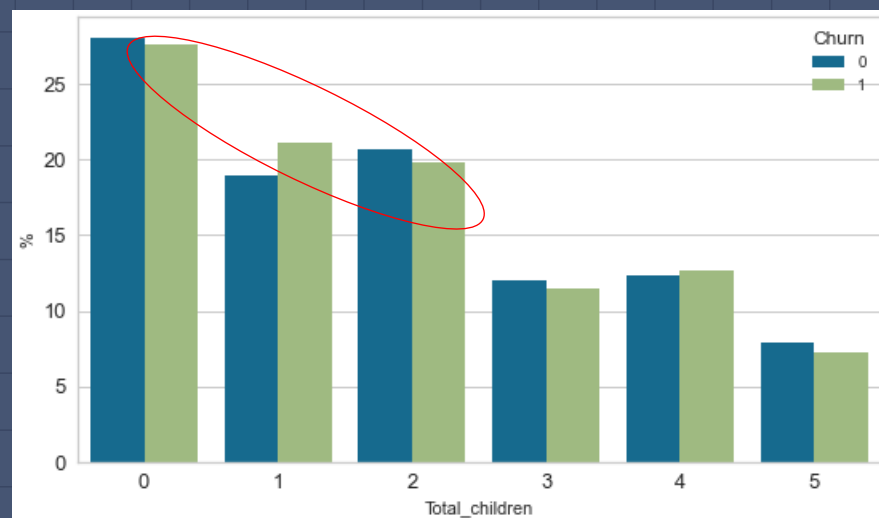
3. CHURN PREDICTIVE MODEL

Data Exploration Analysis

Churn by Number of Children at home



Churn by Total Children



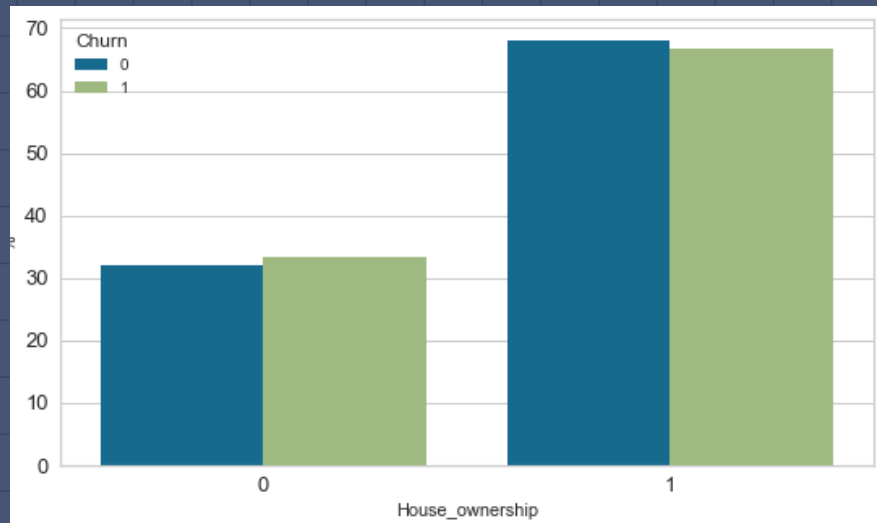
- ❑ Customers with 0 Children at home are the most likely to churn.
- ❑ Customers with 0 Total Children are most likely to churn, but customers with 1 & 2 Children also have a high likelihood of churning.



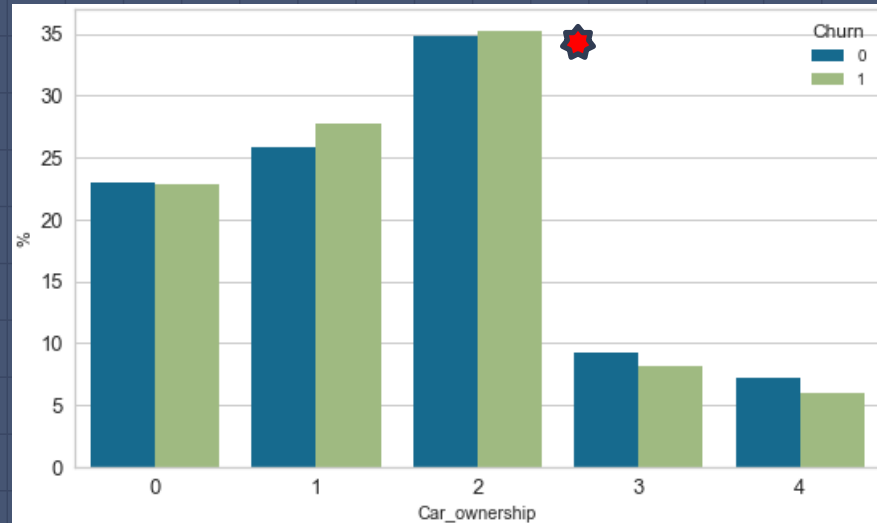
3. CHURN PREDICTIVE MODEL

Data Exploration Analysis

Churn by Home Ownership



Churn by Car Ownership



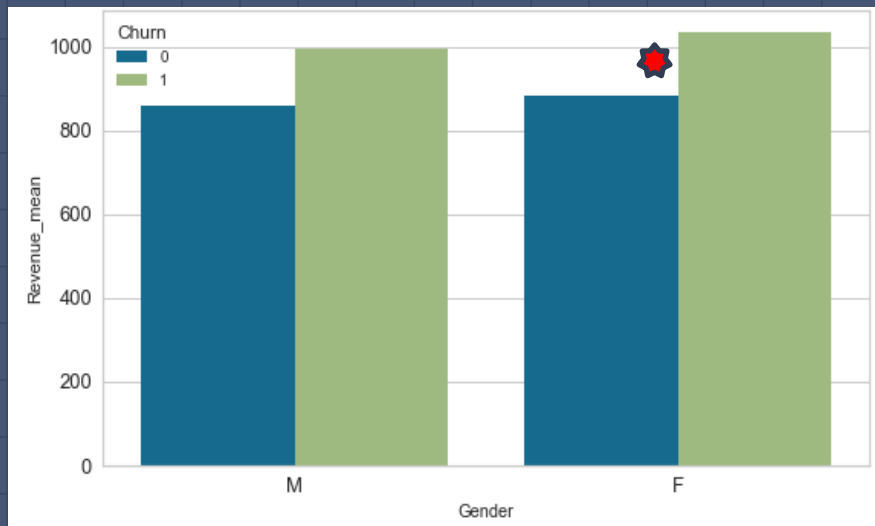
- ❑ No variance for churning with home ownership but customers who own houses are most likely to churn.
- ❑ Customers with 2 cars are most of the customers and have the most churners.



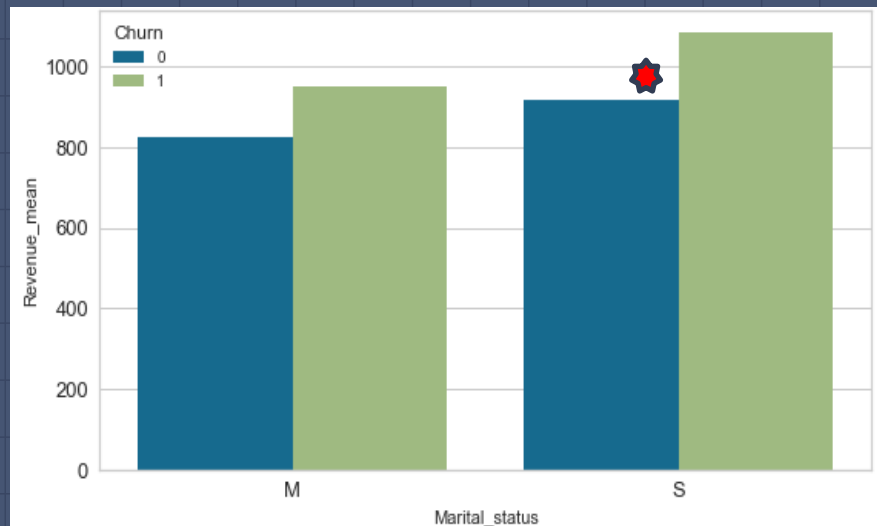
3. CHURN PREDICTIVE MODEL

Data Exploration Analysis

Churn by Gender {Revenue}



Churn by Marital Status {Revenue}



- ❑ In general, Customers who have spent more on average are more likely to churn for Gender and Marital status.
- ❑ Female churners spend slightly more than Male churners. Single Churners spend **much** more than Married Churners by approximately \$150 more on average.



3. CHURN PREDICTIVE MODEL

RFM Analysis

- RFM is important for understanding customer behaviour. The advantages of RFM is that it is easy to implement and it can be used for different types of business. It helps craft better marketing campaigns and improves CRM and customer's loyalty.

- Recency:** How recent is the last purchase of the customer.
- Frequency:** How often the customer makes a purchase.
- Monetary:** How much money does the customer spends.

- RFM scores were applied as follow to the customer single view dataset as follows; with high recency rank (5) meaning low recency “value” and high frequency/monetary rank(5) meaning high frequency/monetary “value”



Quintile of data range	<=20%	21-40%	41-60%	61-80%	>80%
Recency “R” rank	5	4	3	2	1
Frequency “F” rank	1	2	3	4	5
Monetary “M” rank	1	2	3	4	5



3. CHURN PREDICTIVE MODEL

RFM Customer Segmentation

- 10 segments from Recency and Frequency ranks can be created to describe customer behaviour



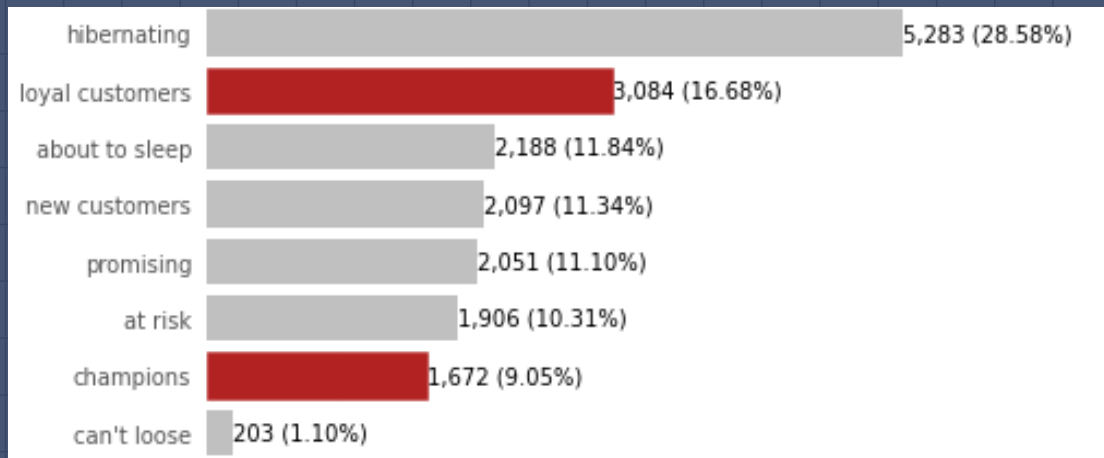
Segment	Description
Champions	Bought recently, buy often and spend the most
Loyal Customers	Buy on a regular basis. Responsive to promotions.
Potential Loyalist	Recent customers with average frequency.
New Customers	Bought most recently, but not often.
Promising	Recent shoppers, but haven't spent much.
Need Attention	Above average recency, frequency and monetary values. May not have bought very recently though.
About To Sleep	Below average recency and frequency. Will lose them if not reactivated.
Can't Lose Them	Used to purchase frequently but haven't returned for a long time.
At Risk	Purchased often but a long time ago. Need to bring them back!
Hibernating	Last purchase was long back and low number of orders. May be lost.



3. CHURN PREDICTIVE MODEL

RFM Customer Segmentation

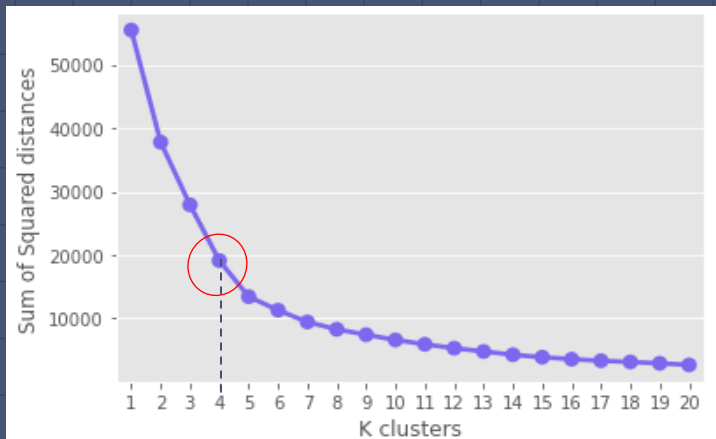
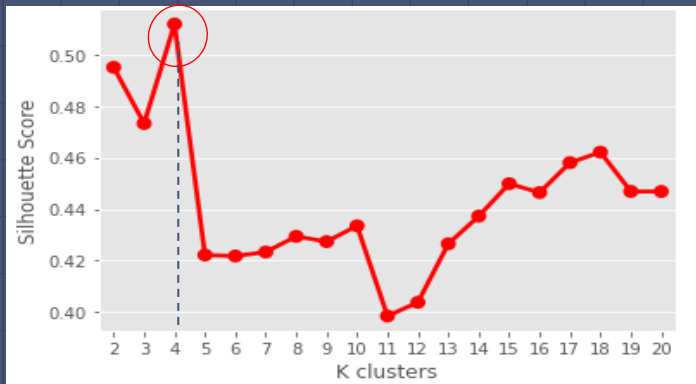
- ❑ **25.7%** of total customers are among AHG's most valuable consumers for the business ("Loyal customers" and "Champions")
- ❑ **10.3%** of customers are "at risk" of being lost and potentially churning/ or churned
- ❑ **28.6%** of customers are "hibernating" and so have the highest propensity for churn out of any other segment.





3. CHURN PREDICTIVE MODEL

RFM Clustering

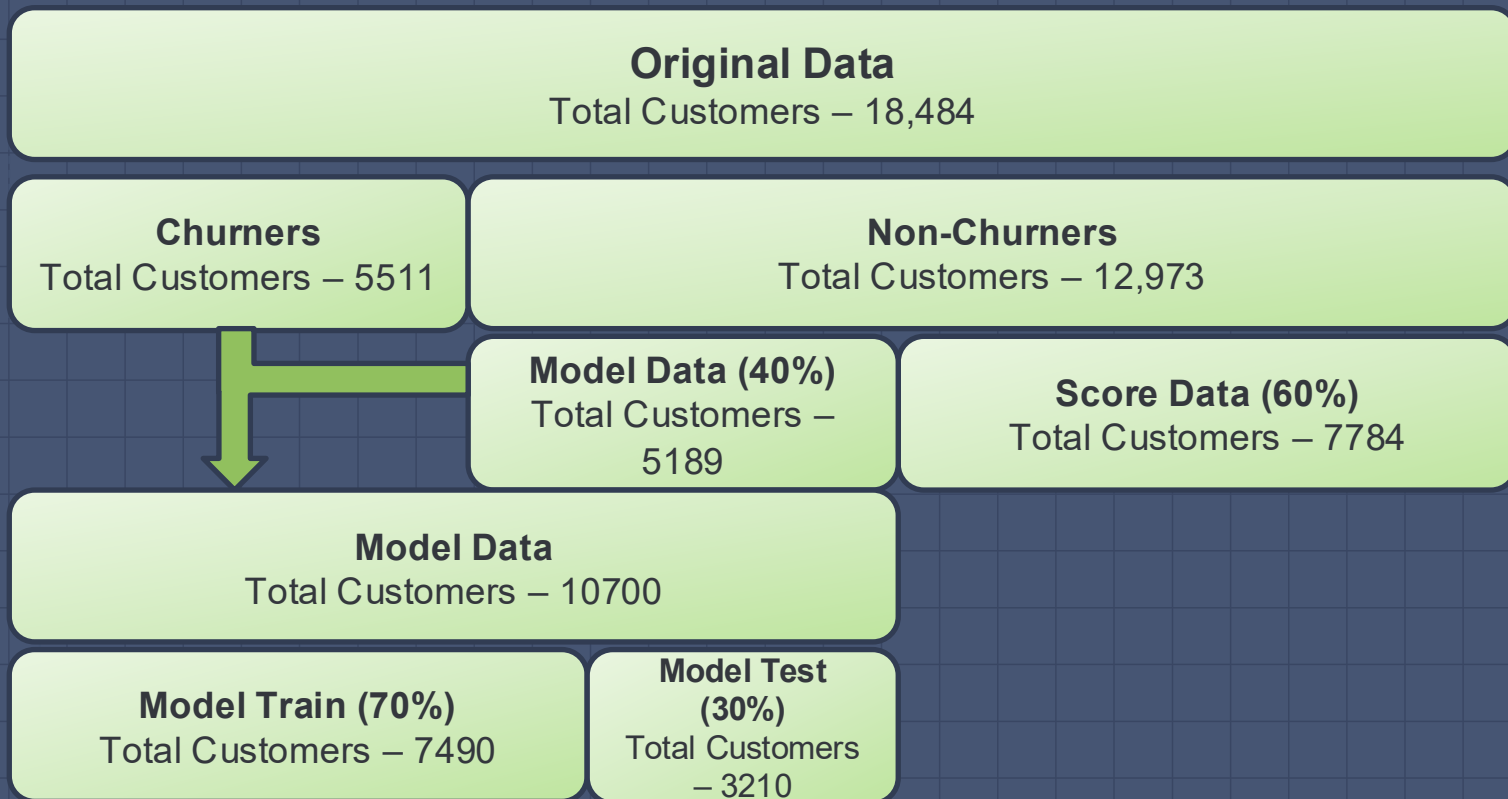


- ❑ Hopkins Statistic was applied to estimate clustering tendency. A value of 0.99 was found, indicating high cluster tendency
- ❑ Both the Silhouette score and Sum of squares(elbow plot) method were used to estimate the optimal K clusters for the K-mean algorithm
- ❑ Both estimated **4 clusters** was optimal
- ❑ A new “RFM_Cluster” field with 4 clusters was added to the single view customer data set for model development.



3. CHURN PREDICTIVE MODEL

Data Selection for model/scoring





3. CHURN PREDICTIVE MODEL

Feature Selection

1. Deleting Irrelevant features

- ❑ Days elapsed aggregated variables contained multiple missing values per customer, therefore needed to be dropped
- ❑ Model Data needs to be trained without customer_ID bias, and therefore is dropped.
- ❑ RFM Score, Recency, Frequency and Monetary variables are removed to remove bias and unsuspected multicollinearity between the variables and “RFM_Cluster”

Deleted Features	
Customer_ID	Order Quantity_min
Birth_Date	Days Elapsed_sum
Days Elapsed_mean	Days Elapsed_median
Days Elapsed_min	Days Elapsed_max
RFM Score	Recency
Monetary	Frequency



3. CHURN PREDICTIVE MODEL

Feature Selection

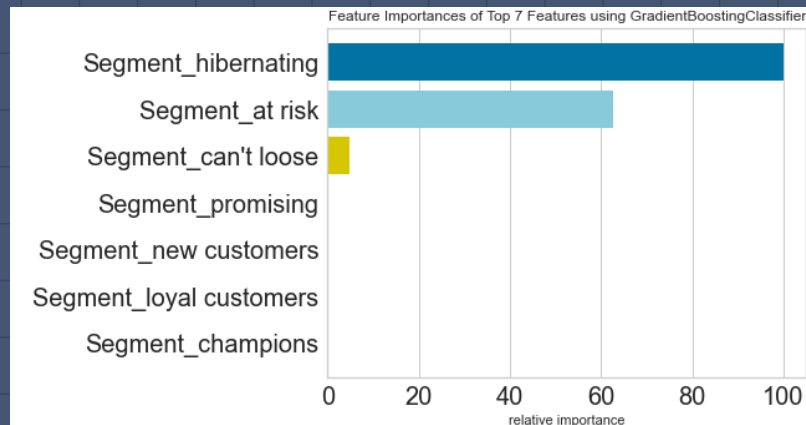
2. Encoding Categorical Data

- ❑ In addition to the attitudinal sales reason data, other categorical features were encoded before the selection

Encoded Categorical Features	
Gender	Marital_status
Education	Occupation
Segment	Commute_Distance
Age Group	Income Group
Sales Reason	Sales Reason Type

3. Feature Importance

- ❑ Using “Feature Importance”, only the most important rfm segment will be added to boost performance and still maintain controlled collinearity from other rfm variables (rfm clusters)
- ❑ Most important feature to add is “Segment_hibernating”. All others will be dropped.





3. CHURN PREDICTIVE MODEL

Feature Selection

4. Voting Selector

- ❑ Using Xverse Voting Selector to apply 6 different feature selection methods. Each feature is given a score for each of the 6 methods they're successfully chosen in
- ❑ Features were filtered by scores > 4

Xverse's 6 feature selection methods

- Information Value using Weight of evidence
- Variable Importance using Random Forest.
- Recursive Feature Elimination.
- Variable Importance using Extra trees classifier.
- Chi-Square best variables.
- L1 based feature selection.

5. Multicollinearity (with VIF)

- ❑ The Final features were selected by using Variable inflation factors (VIF) on the remaining features
- ❑ For no strong multicollinearity, the VIF for all the variables must be **below 5**

Final Features

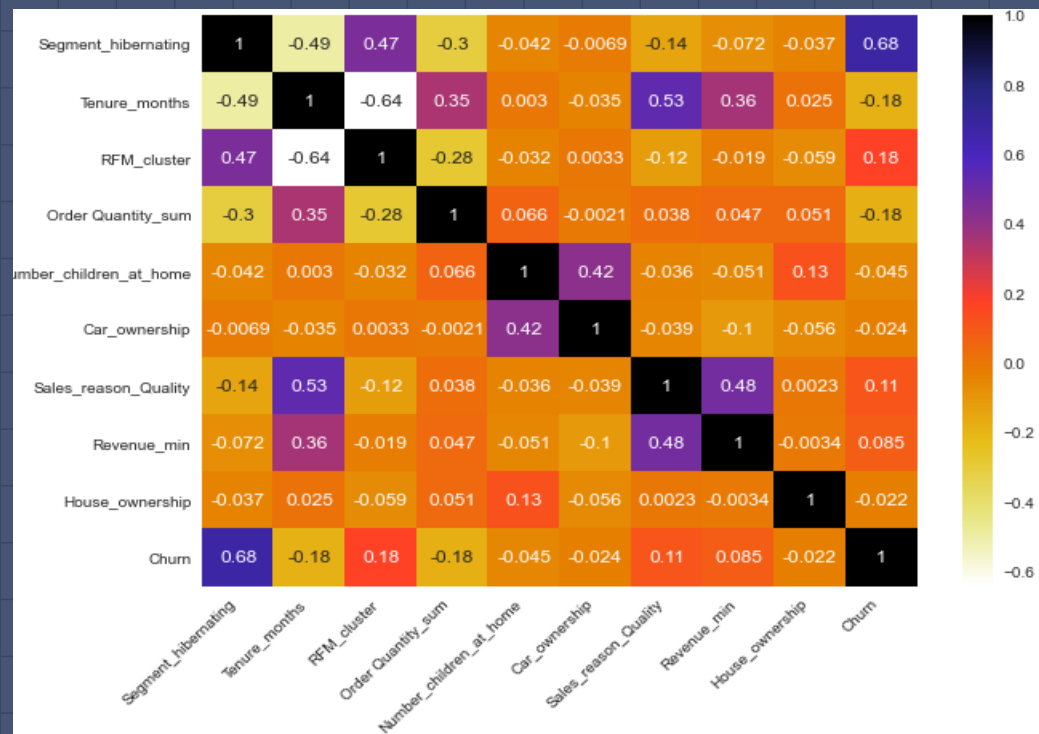
Segment_hibernating	Tenure_months
RFM_Cluster	Order Quantity_Sum
Number_children_at_home	Car_ownership
Sales_reason_Quality	Revenue_min
House_ownership	



3. CHURN PREDICTIVE MODEL

Feature Selection

6. Pearson's correlation



- ❑ The final features are evaluated on correlation heatmap
- ❑ None of the variables have a high correlation for one another (≥ 0.8)



3. CHURN PREDICTIVE MODEL

Model Selection

**Logistic
Regression**

**Linear
Discriminant
Analysis**

**Gradient
Boosting**

**K Nearest
Neighbours**

Random Forest

Extra Trees

Decision Tree

Naïve Bayes

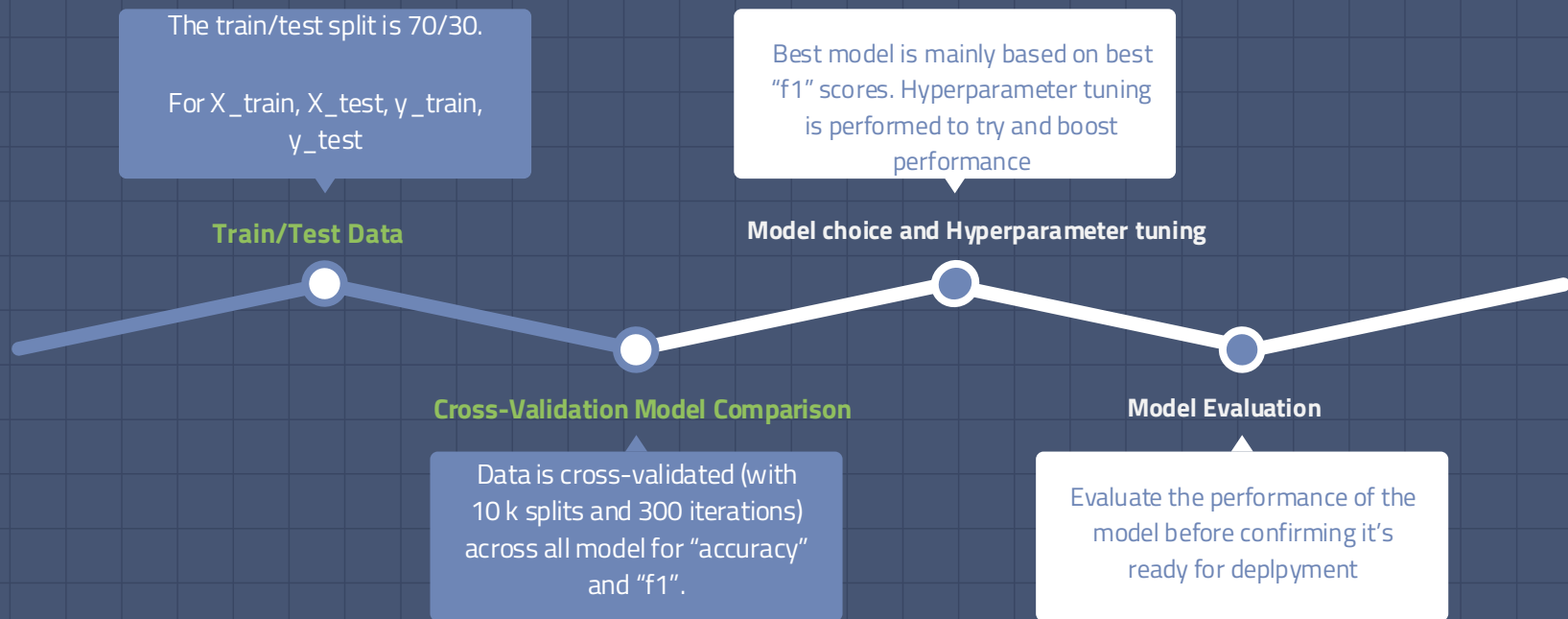
**Support Vector
Machines**

Neural Network



3. CHURN PREDICTIVE MODEL

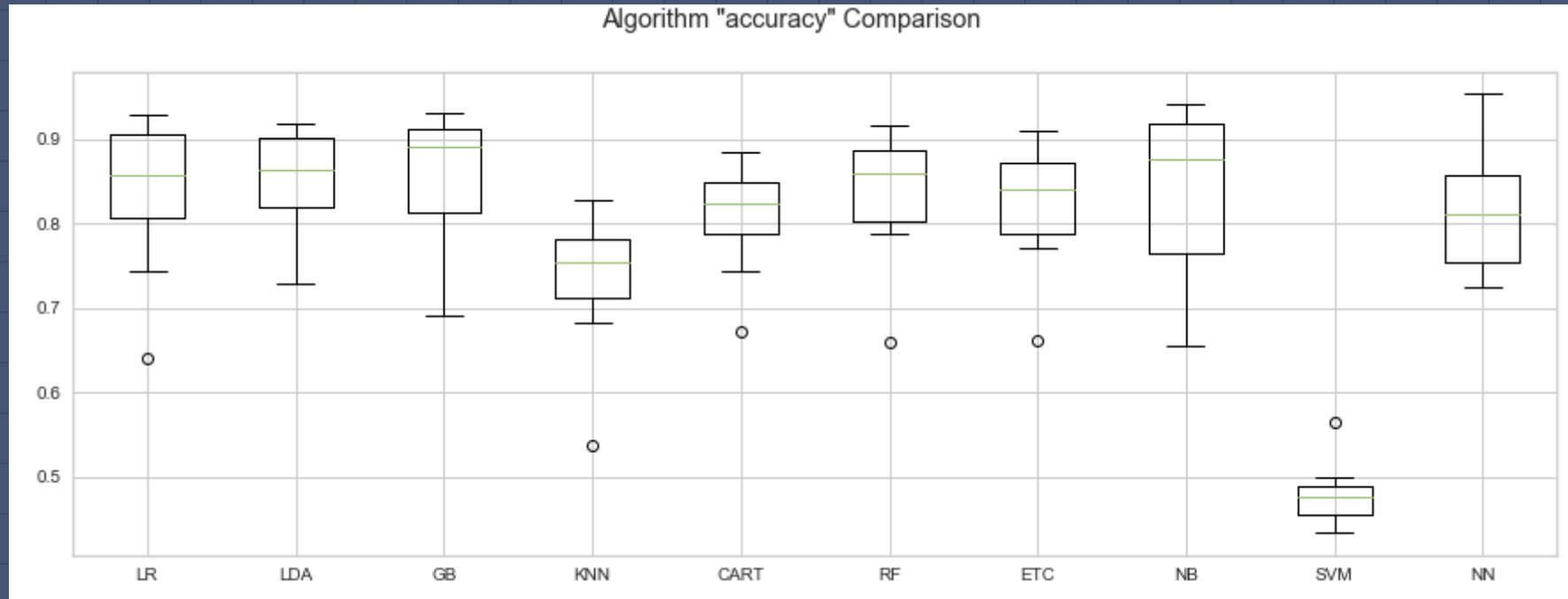
Model Selection



3. CHURN PREDICTIVE MODEL



Model Selection

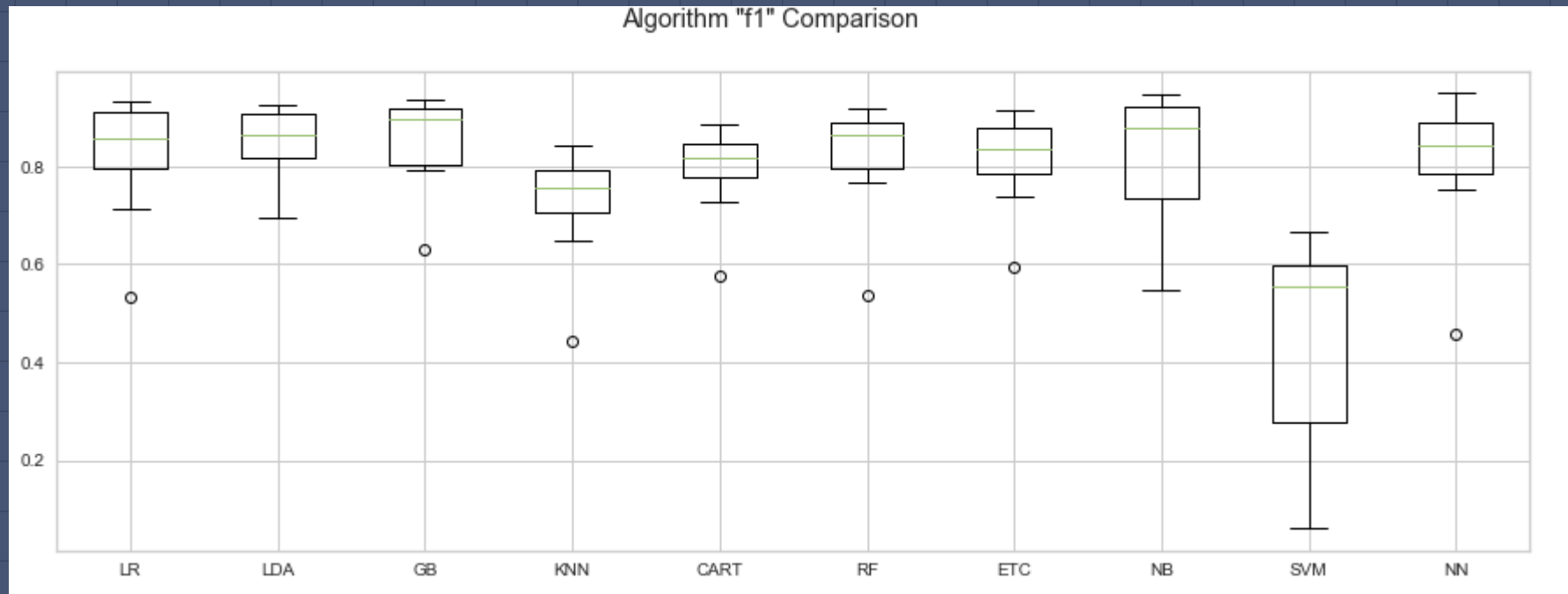


Note - Better scaling of final model data could improve performance of SVM and NN*

3. CHURN PREDICTIVE MODEL



Model Selection



Note- Better scaling of final model data could improve performance of SVM and NN*



3. CHURN PREDICTIVE MODEL

Model Selection

Model	Accuracy %	Std %	F1 Score %	Std %
Logistic Regression	83.60	8.50	82.32	11.58
Linear Discriminant Analysis	84.79	6.55	84.55	7.88
Gradient Boosting	85.82	7.23	85.31	9.03
K Neighbours	73.28	7.69	72.49	10.78
Decision Tree	80.90	6.15	79.64	8.68
Random Forest	83.69	7.27	82.39	10.68
Extra Trees	82.42	6.99	81.52	8.98
Gaussian NB	83.85	9.31	81.94	12.54
Support Vector Machines	47.75	3.48	44.45	22.28
Neural Networks	81.29	6.76	81.29	13.25

- ❑ Linear Discriminant Analysis (LDA) chosen as the best model algorithm
- ❑ Although, it comes 2nd after gradient boosting in mean f1 score. It has a much smaller “std%” thus meaning it will consistently reach better f1 levels.

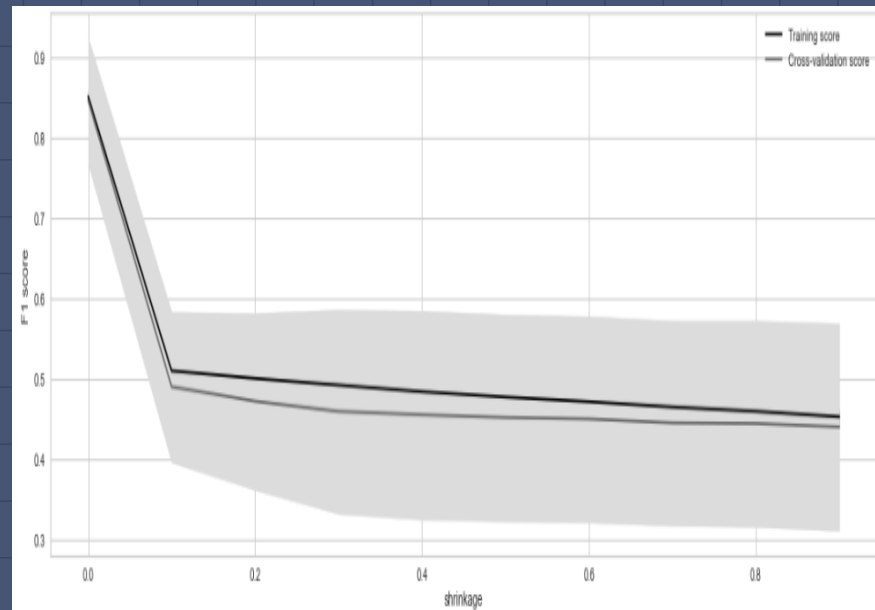




3. CHURN PREDICTIVE MODEL

Hyperparameter tuning

- ❑ LDA was tested on 3 different solvers; “svd”, “lsq” and “eigen”. The svd solver (which was already default) had the best f1 scores
- ❑ Shrinkage was also tuned during the testing. As shown in the validation curve, any value of shrinkage above 0, lead to an immediate decrease in performance
- ❑ Therefore, the default LDA parameters were already the best parameters for the model.





3. CHURN PREDICTIVE MODEL

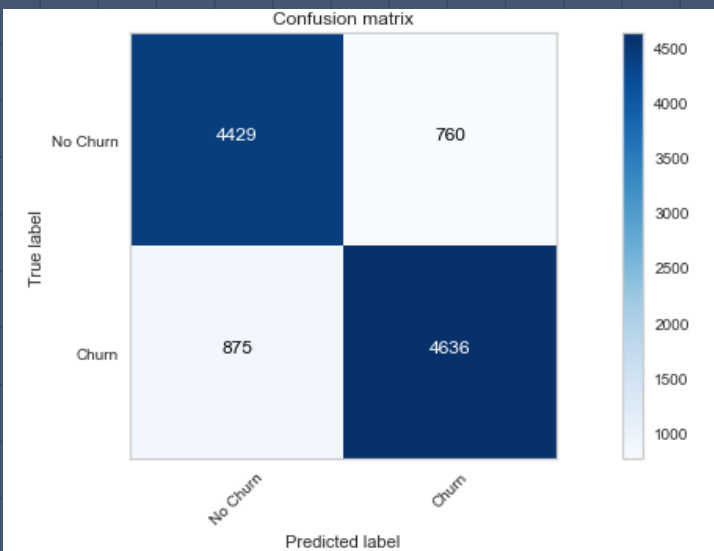
Model Evaluation

	precision	recall	f1-score	support
0	0.84	0.85	0.84	5189
1	0.86	0.84	0.85	5511
accuracy			0.85	10700
macro avg	0.85	0.85	0.85	10700
weighted avg	0.85	0.85	0.85	10700

Classification Report

- ❑ The average Precision, Recall, accuracy and f1 score are 85% (0.85)

Confusion Matrix



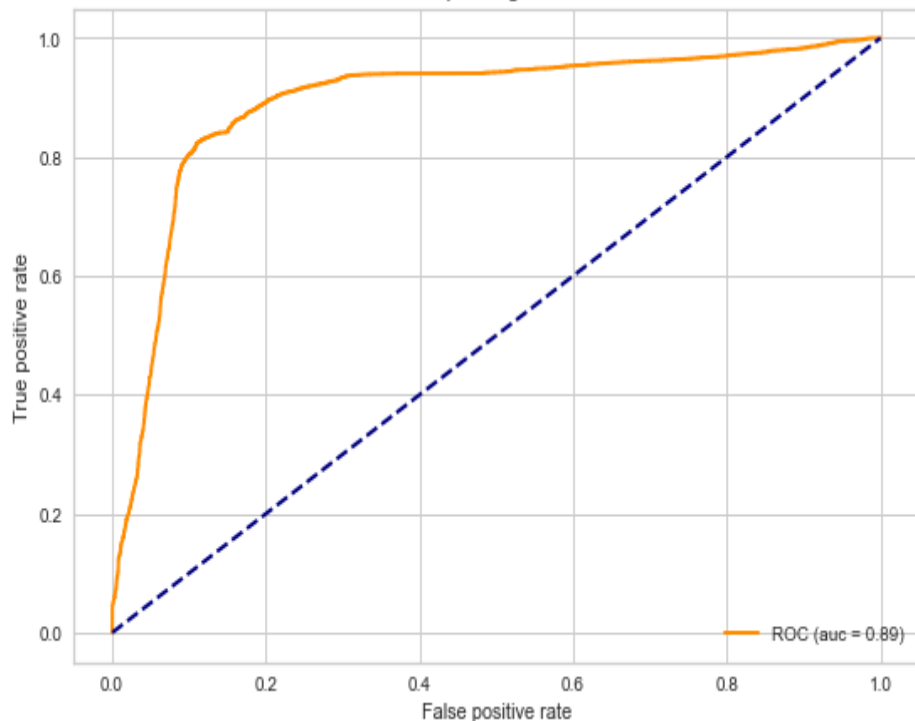
- ❑ Customers classified correctly as Non Churners – 4429
- ❑ Customers classified correctly as Churners – 4636
- ❑ Customers classified incorrectly as Non - Churners – 875
- ❑ Customers classified incorrectly as Churners - 760



3. CHURN PREDICTIVE MODEL

Model Evaluation

Receiver operating characteristic



ROC curve (Area under the curve)

- ❑ AUC of 0.89 indicates a very good model

3. CHURN PREDICTIVE MODEL



Model deployment on score data + impact

Score Data	
Total Customers	7784 customers
Total predicted Churners	1185 churners
Total predicted Non-churners	6599 non-churners
Average revenue per customer per year-	\$ 1975

If the strategy is successful is successful in preventing churning
Then AHG can potentially save **\$2,340,783 !**

3. CHURN PREDICTIVE MODEL



Recommendations to prevent churn model

- ❑ A/B testing proved the null hypothesis that AHG management targeting only predicted churners with marketing campaign will maximise profits (due to more controlled customer retention)
- ❑ **Identify customers that are likely to churn** on a weekly basis. Any customer that has not purchased items over the last 5 months is to be classified as a churner or non-churner
- ❑ Upon identifying a potential churner, **email/post a promotional voucher or discount coupon**
- ❑ **Identify recommended product from the recommendation model** and send related promotions to the buyer
- ❑ Ask for feedback and recommendations on product range + specification from customer to **engage and revive them**

THANK
YOU

